(b) What is the center of A?

Hint: Show that x^p and y^p are central elements.

(c) Find all irreducible finite dimensional representations of A.

Hint: Let V be an irreducible finite dimensional representation of A, and let v be an eigenvector of y in V. Show that the collection of vectors $\{v, xv, x^2v, \ldots, x^{p-1}v\}$ is a basis of V.

Problem 2.7.5. Let q be a nonzero complex number, and let A be the q-Weyl algebra over $\mathbb{C}$.

(a) What is the center of A for different q? If q is not a root of unity, what are the two-sided ideals in A?

(b) For which q does this algebra have finite dimensional representations?

Hint: Use determinants.

(c) Find all finite dimensional irreducible representations of A for such q.

Hint: This is similar to part (c) of the previous problem.

2.8. Quivers

Definition 2.8.1. A **quiver** Q is a directed graph, possibly with self-loops and/or multiple edges between two vertices.

Example 2.8.2.

$$\begin{tikzcd} \bullet \arrow[r] & \bullet & \bullet \arrow[l] \\ & \bullet \arrow[u] & \end{tikzcd}$$

We denote the set of vertices of the quiver Q as I and the set of edges as E. For an edge $h \in E$, let h', h'' denote the source and target of h, respectively:

$$\underset{h'}{\bullet} \xrightarrow{h} \underset{h''}{\bullet}$$

Definition 2.8.3. A **representation of a quiver** Q is an assignment to each vertex $i \in I$ of a vector space V_i and to each edge $h \in E$ of a linear map $x_h : V_{h'} \longrightarrow V_{h''}$.